Fizika

1. Uzunligi 60 sm bo'lgan ipga osilgan yuk tekis harakatlanib, gorizontal tekislikda aylana chizadi. Yuk harakatlanayotgan vaqtda ip vertikal bilan 30^0 o'zgarmas burchak tashkil qilsa, yuk qanday tezlik bilan harakatlanayotgan bo'ladi?

Berilgan: $l = 0{,}6\ m$, $\alpha = 30^0$, $g = 9{,}8\ m/s^2$;
Topish kerak: $\upsilon - ?$
Masalaning yechilishi:

Jismga og'irlik kuchi hamda ipning taranglik kuchi ta'sir etadi. Jism harakati uchun dinamikaning asosiy tenglamasini vektor ko'rinishida yozamiz:

$$m\bar{g} + \bar{T} = m\bar{a} \qquad (1)$$

Ox o'qini gorizontal, *Oy* o'qini unga tik yo'naltirib, (1) ifodaga kirgan vektor kattaliklarni mazkur o'qlarga proeksiyalari orqali uni qayta yozamiz:

Ox o'qiga: $T\sin\alpha = ma$ (2)
Oy o'qiga: $-mg + T\cos\alpha = 0$ (3)

Markazga intilma tezlanish ifodasi $a = \upsilon^2 / r$, bu yerda r–jism harakatlanayotgan aylana radiusi. 1,1 –rasmdan $r = l\sin\alpha$. Bularni e'tiborga olib, (2) va (3) ifodalarning nisbatini olamiz:

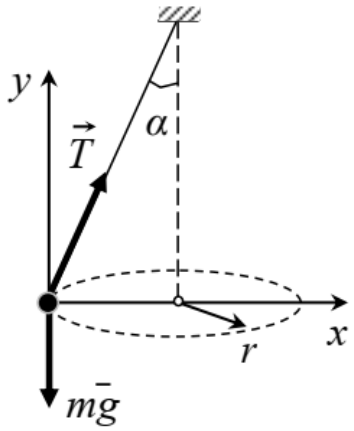
$$\frac{\upsilon^2}{gl\sin\alpha} = tg\alpha \qquad (4)$$

Bundan yakuniy ifoda kelib chiqadi:

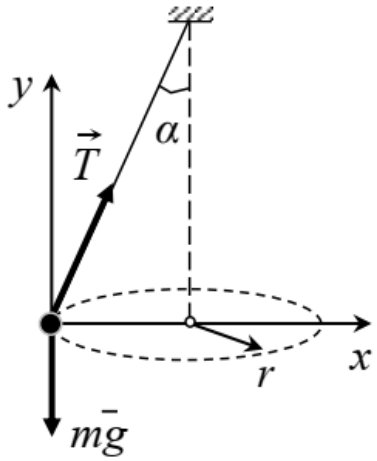
$$\boxed{\upsilon = \sqrt{g\,l\sin\alpha\,tg\alpha}} \qquad (5)$$

υ ning o'lchov birligini tekshiramiz: $[\upsilon] = \sqrt{\frac{m}{s^2}\cdot m} = \frac{m}{s}$;

Hisoblash natijasi: $\upsilon = 1{,}31\ m/s$.



1,1-rasm



2,1-rasm

2. Massalari 0,7 kg va 1,3 kg bo'lgan yuklar ip orqali qo'zg'almas blokka ilingan. Dastlab yuklar bir xil balandlikda harakatsiz ushlab turilibdi.

Berilgan: $m = 45\ kg,\ l = 5\ m,\ \nu = 0{,}27\ s^{-1},\ g = 9{,}8\ m/s^2;$
Topish kerak: $\alpha - ?\quad T - ?$
Masalaning yechilishi:

Jismga og'irlik kuchi hamda ipning taranglik kuchi ta'sir etadi. Jism harakati uchun dinamikaning asosiy tenglamasini vektor ko'rinishida yozamiz:

$$m\bar{g} + \bar{T} = m\bar{a} \qquad (1)$$

Ox o'qini gorizontal, *Oy* o'qini unga tik yo'naltirib, (1) ifodaga kirgan vektor kattaliklarni mazkur o'qlarga proeksiyalarini olamiz:

Ox o'qiga: $T\sin\alpha = ma$ (2)
Oy o'qiga: $-mg + T\cos\alpha = 0$ (3)

Markazga intilma tezlanish ifodasi $a = 4\pi^2\nu^2 r$, bu yerda r–jism harakatlanayotgan aylana radiusi. 2,1 –rasmdan $r = l\sin\alpha$. Bu ifodalarni e'tiborga olib, (2) va (3) dan izlanilayotgan kattaliklarni hisoblash tenglamalarini yozish mumkin:

$$\boxed{\alpha = \arccos(g/4\pi^2\nu^2 l)} \qquad (4)$$

$$\boxed{T = 4\pi^2\nu^2 l\, m} \qquad (5)$$

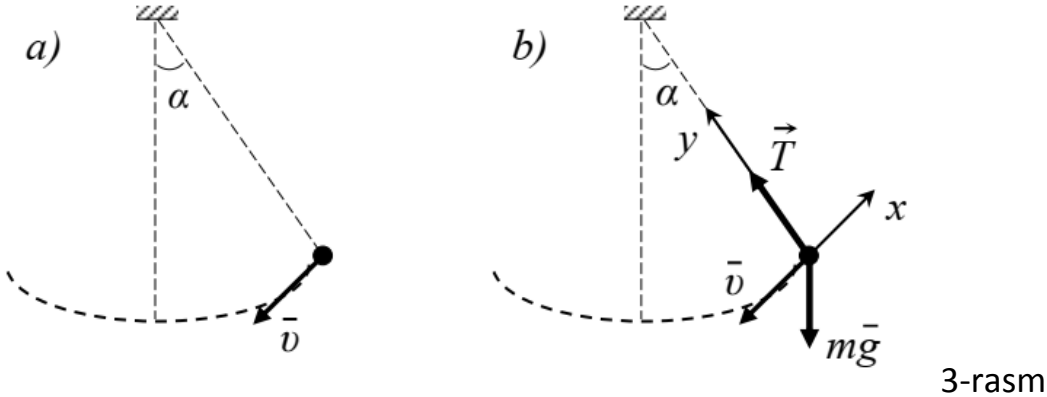
T ning o'lchov birligini tekshiramiz: $[T]=\frac{kg\cdot m}{s^2}=N$;

Hisoblash natijalari: $\alpha=45^0,\ T=0{,}63\ kN$.

3. Agar 3–rasm (*a*) da ko'rsatilgan ipga osilgan yukning massasi 100 g, tezligi 2 m/s, burchak 60^0 bo'lsa, ipning rasmda ko'rsatilgan paytdagi taranglik kuchini toping. Ipning uzunligi 40 sm.

Berilgan: $m=0{,}1\ kg,\ \upsilon=2\ m/s,\ \alpha=60^0,\ l=0{,}4\ m$;
Topish kerak: $T-?$
Masalaning yechilishi:

Jismga og'irlik kuchi hamda ipning taranglik kuchi ta'sir etadi. Jism harakati uchun dinamikaning asosiy tenglamasini vektor ko'rinishida yozamiz:



3-rasm

$$m\bar{g}+\bar{T}=m\bar{a} \qquad (1)$$

(1) ifodaga kirgan vektor kattaliklarni Oy o'qiga (3–rasm *b*) proeksiyasini olamiz:

$$-mg\cos\alpha+T\sin\alpha=ma \qquad (2)$$

Aylanma harakat kinematikasidan markazga intilma tezlanish: $a=\frac{\upsilon^2}{l}$. Bu tenglikni (2) ga qo'yamiz, undan masalaning ishchi formulasi kelib chiqadi:

$$\boxed{T=mg\cos\alpha+m\frac{\upsilon^2}{l}} \qquad (3)$$

T ning o'lchov birligini tekshiramiz: $[T] = kg \cdot \frac{m}{s^2} = N;$

Hisoblash natijasi: $T = 1{,}5\ N.$

@olimpdep

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